Data Communications & Networks

Chapter 6: Sections 6.1 & 6.2

Bandwidth Utilization: Multiplexing & Spread Spectrum

Topics Covered:

- 6.1.1 Frequency-Division Multiplexing (FDM)
- 6.1.2 Wavelength-Division Multiplexing (WDM)
- 6.1.3 Time-Division Multiplexing (TDM): Sync & Statistical
- 6.2.1 Frequency Hopping Spread Spectrum (FHSS)
- 6.2.2 Direct Sequence Spread Spectrum (DSSS)

Reference: Forouzan, "Data Communications and Networking," 5th ed.

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Contents

1 Multiplexing — Section 6.1 **2**

1.1 Frequency-Division Multiplexing (FDM) 2

1.1.1 Core Concept 3

1.1.2 Numerical Examples 3

1.1.3 Analog Telephone Hierarchy 5

1.2 Wavelength-Division Multiplexing (WDM) 5

1.3 Time-Division Multiplexing (TDM) 6

1.3.1 Core Concept 6

1.3.2 Synchronous TDM 6

1.3.3 Digital Signal (DS) / T-Line Hierarchy 8

1.3.4 Statistical TDM 9

2 Spread Spectrum — Section 6.2 **10**

2.1 Frequency Hopping Spread Spectrum (FHSS) 10

2.1.1 How FHSS Works 10

2.1.2 FHSS Bandwidth 11

2.2 Direct Sequence Spread Spectrum (DSSS) 12

2.2.1 How DSSS Works 12

2.2.2 Comparison: FHSS vs DSSS 13

Chapter 1 Multiplexing — Section 6.1

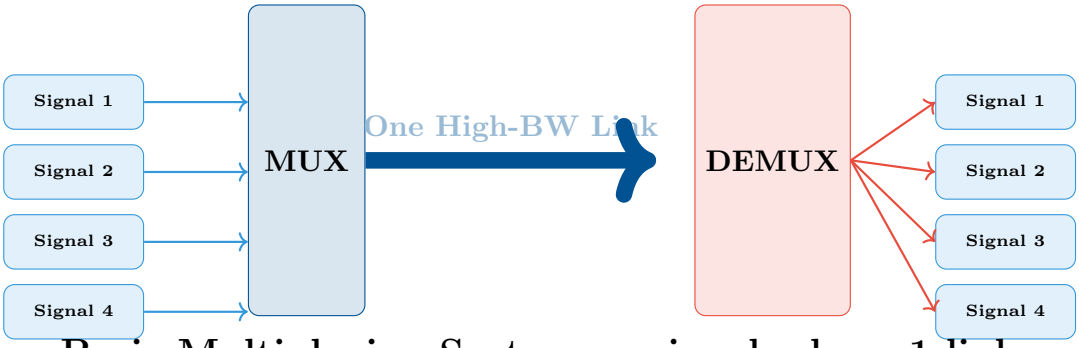
The Problem Multiplexing Solves

Imagine you own an expensive 8-lane highway (high-bandwidth fiber cable), but you only need to send one car (one signal). That's wasteful! **Multiplexing** is the engineering solution: it lets multiple signals *share* one high-bandwidth channel simultaneously.

Why do we need multiplexing?

- Installing a separate cable for each pair of users is astronomically expensive.
- Today's transmission media (fiber optics, satellite) have enormous bandwidth — far more than any single signal needs.
- Multiplexing packs many signals onto one link, maximizing efficiency and slashing cost.

In a multiplexed system, n signals share the bandwidth of one link. A **multiplexer (MUX)** combines them at the sender side; a **demultiplexer (DEMUX)** splits them at the receiver side.



Basic Multiplexing System: n signals share 1 link

1.1 Frequency-Division Multiplexing (FDM)

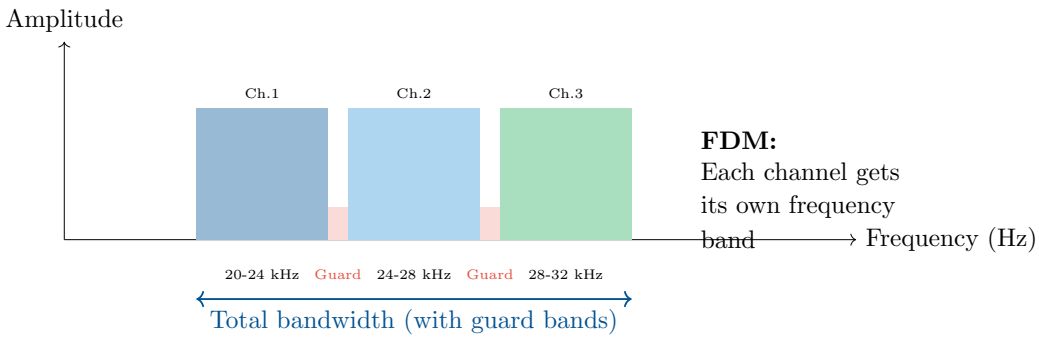
1.1.1 Core Concept

FDM: Splitting the Frequency Spectrum

FDM is an *analog* technique. Think of it like a radio dial: many stations broadcast simultaneously on the same air, each on a *different frequency*. You tune to one frequency to get one station.
How it works:

1. Each input signal occupies a **base bandwidth** (e.g., 4 kHz for voice).
2. Each signal is **modulated** onto a different carrier frequency: $f_1, f_2, f_3, \ldots$
3. The modulated signals are **combined** (added together) into one composite signal.
4. This composite signal is sent over the shared link.
5. At the receiver, **bandpass filters** extract each individual signal.
6. Each filtered signal is **demodulated** to recover the original.

Guard bands: Thin strips of unused spectrum are placed between channels to prevent them from overlapping (interference).



1.1.2 Numerical Examples

Example 6.1: Three 4-kHz Voice Channels via FDM

Given: 3 voice channels, each with 4 kHz bandwidth. Available link: 20–32 kHz. No guard bands. Show the FDM configuration.

Solution:

- Total available bandwidth: $32 - 20 = 12$ kHz.
- Each channel needs 4 kHz $\Rightarrow 3 \times 4 = 12$ kHz. Perfect fit!
- Channel 1: modulated to occupy **20–24 kHz** (carrier $f_1 = 22$ kHz center)
- Channel 2: modulated to occupy **24–28 kHz** (carrier $f_2 = 26$ kHz center)
- Channel 3: modulated to occupy **28–32 kHz** (carrier $f_3 = 30$ kHz center)

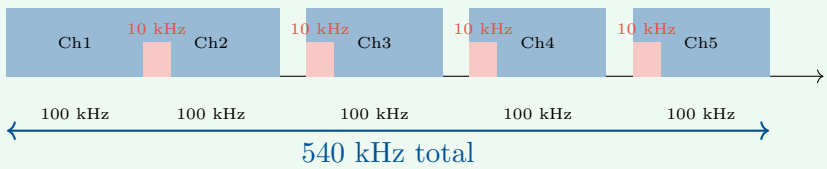
At the receiver: filter 1 passes 20–24 kHz, filter 2 passes 24–28 kHz, filter 3 passes 28–32 kHz. Each filtered signal is then shifted back to 0–4 kHz by demodulation.

Example 6.2: Guard Bands Required

Given: 5 channels, each with 100 kHz bandwidth. Guard band = 10 kHz between channels. Find the minimum total link bandwidth.

Solution:
With 5 channels, there are **4 gaps** between them (between channels 1–2, 2–3, 3–4, 4–5):

$$\text{Total BW} = 5 \times 100 \text{ kHz} + 4 \times 10 \text{ kHz} = 500 + 40 = \mathbf{540} \text{ kHz}$$



Example 6.3: Digital Sources + FDM via Modulation

Given: 4 digital channels, each at 1 Mbps, using a satellite analog channel of 1 MHz total. Design an FDM configuration.

Solution:

- Divide the 1 MHz satellite channel into 4 sub-channels of $\frac{1000}{4} = \mathbf{250}$ kHz each.
- Each 1 Mbps digital source is modulated using **16-QAM** (which encodes 4 bits per symbol).
- With 16-QAM, 1 Mbps digital data needs: $\frac{1 \text{ Mbps}}{4 \text{ bits/symbol}} = 250{,}000$ baud $=$ 250 kHz. Fits perfectly!

Example 6.4: AMPS Cellular FDM (Real World)

Given: AMPS uses band 824–849 MHz (sending) and 869–894 MHz (receiving). Each user gets a 30 kHz channel. How many simultaneous users?

Solution:

$$\text{Band width} = 849 - 824 = 25 \text{ MHz}$$

$$\text{Number of channels} = \frac{25 \text{ MHz}}{30 \text{ kHz}} = \frac{25{,}000 \text{ kHz}}{30 \text{ kHz}} = 833.3 \approx 832 \text{ channels}$$

Of 832 channels, 42 are for control $\Rightarrow$ **790 channels for voice users**.

1.1.3 Analog Telephone Hierarchy

The telephone system uses a structured FDM hierarchy to combine thousands of calls:

Level	Name	Bandwidth	Voice Channels	Composition
Base	Voice	4 kHz	1	—
1	Group	48 kHz	12	12 voice
2	Supergroup	240 kHz	60	5 groups
3	Master group	2.52 MHz	600	10 supergroups
4	Jumbo group	16.984 MHz	3600	6 master groups

Real-World FDM Applications:

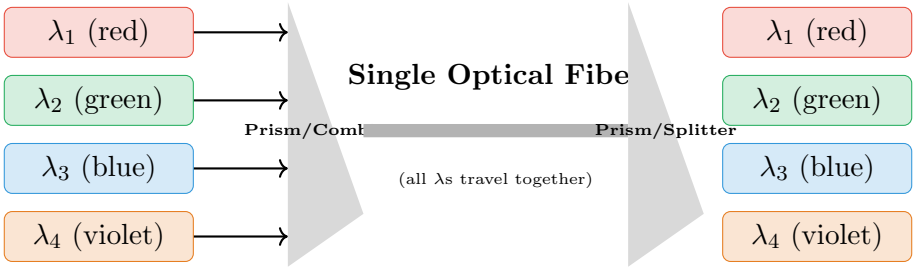
- **AM radio**: 530–1700 kHz band; each station uses 10 kHz.
- **FM radio**: 88–108 MHz band; each station uses 200 kHz.
- **TV channels**: Each channel has 6 MHz bandwidth.
- **1st-gen cellular (AMPS)**: 30 kHz per user channel (FDM in air).
- **DSL**: Splits phone line into voice (low frequencies) and data (high frequencies).

1.2 Wavelength-Division Multiplexing (WDM)

WDM: FDM for Light

WDM is essentially FDM for **optical fiber**. Instead of using different radio frequencies, WDM uses different **wavelengths of light** (colors). Each "color" carries a separate signal.

- Optical prisms (or diffraction gratings) combine and separate different light wavelengths.
- Each wavelength can carry a very high data rate (e.g., 10 Gbps each).
- **DWDM (Dense WDM)**: Can multiplex 32, 64, or even 160 wavelengths on one fiber.
- Used in long-haul internet backbone (trans-oceanic cables).



1.3 Time-Division Multiplexing (TDM)

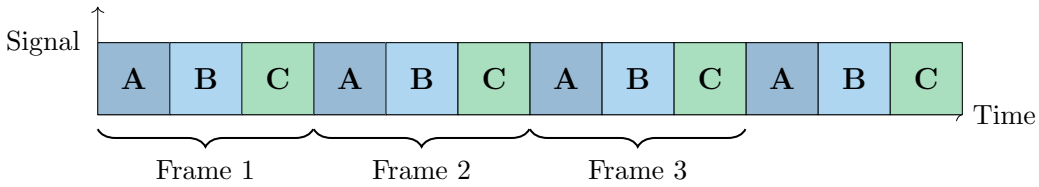
1.3.1 Core Concept

TDM: Sharing Time Instead of Frequency

While FDM divides the frequency spectrum, **TDM divides time**. Each input signal gets the *entire bandwidth of the link*, but only for a brief **time slot**. The multiplexer rapidly cycles through all input channels, giving each a turn on the link. The channel is shared **in time**.

Analogy: One teacher (link) teaches 30 students (channels). In FDM, each student gets their own quiet corner (separate frequency). In TDM, each student gets the teacher's full attention for 2 minutes, then the next student's turn comes. The teacher rotates rapidly enough that everyone is served effectively.

TDM is a **digital** technique — it works with discrete data (bits, characters, blocks).



Each frame = 1 slot for each channel. Channel A, B, C share the link in time.

1.3.2 Synchronous TDM

Synchronous TDM

In **synchronous TDM**:

- Each input gets a **fixed, reserved time slot** in every frame — even if it has no data to send!
- A **frame** = one complete cycle of all n time slots.
- If input unit duration = T s, the output time slot = T/n s (n times shorter).
- Output data rate = $n\times$ input data rate (each of n inputs contributes equally).

Key relationships:

$$\text{Output frame rate} = \text{Input data rate (frames per second)}$$
$$\text{Output bit rate} = n \times \text{Input bit rate}$$
$$\text{Frame duration} = \frac{1}{\text{Frame rate}} = \frac{1}{\text{Input data rate}}$$

Worked Numerical Examples

Example 6.6: Four 1-Mbps channels multiplexed

Given: 4 input channels, each at 1 Mbps. Unit = 1 bit. Find: (1) Input bit duration, (2) Output bit duration, (3) Output bit rate, (4) Frame rate.

Solution:

1. **Input bit duration:** $T_{in} = \frac{1}{1 \text{ Mbps}} = 1\ \mu\text{s}$ per bit.
2. **Output bit duration:** The output slot is $n = 4$ times shorter.
$$T_{out} = \frac{T_{in}}{n} = \frac{1\ \mu\text{s}}{4} = \mathbf{0.25}\ \mu\text{s} = \mathbf{250}\ \text{ns}$$
3. **Output bit rate:** $N_{out} = \frac{1}{T_{out}} = \frac{1}{0.25\ \mu\text{s}} = \mathbf{4}$ Mbps
Or simply: $N_{out} = n \times N_{in} = 4 \times 1$ Mbps $= 4$ Mbps.
4. **Frame rate:** Frame rate = input data rate (each input contributes 1 bit per frame).
$$\text{Frame rate} = 1{,}000{,}000 \text{ frames/s} = \mathbf{1} \text{ Mframes/s}$$
Verify: 1,000,000 frames/s × 4 bits/frame = 4 Mbps. ✓

Example 6.7: Four 1-kbps channels multiplexed

Given: 4 connections at 1 kbps each. Unit = 1 bit.

Solution:

$$\text{Output bit rate} = 4 \times 1 \text{ kbps} = \mathbf{4} \text{ kbps}$$
$$\text{Output bit duration} = \frac{1}{4 \text{ kbps}} = \mathbf{0.25} \text{ ms}$$
$$\text{Frame rate} = 1{,}000 \text{ frames/s (same as input rate)}$$
$$\text{Frame duration} = \frac{1}{1000} = \mathbf{1} \text{ ms}$$

Example 6.8: Byte interleaving — 100 bytes/s per channel

Given: 4 channels, each sending 100 bytes/s. Interleaved unit = 1 byte (8 bits).

Solution:

- Frame size: 4 channels $\times$ 1 byte = **4 bytes = 32 bits per frame**.
- Frame rate: Each channel sends 100 bytes/s, so there are **100 frames/s**.
- Frame duration: $\frac{1}{100}$ = **0.01** s = **10** ms.
- Link bit rate: 100 frames/s $\times$ 32 bits/frame = **3200** bps.
- Check: $4 \times 100 \times 8 = 3200$ bps. ✓

Example 6.9: Four 100-kbps channels with bit interleaving

Given: 4 channels, each at 100 kbps. Unit = 1 bit. Find frame details.

Solution:

- Output bit rate: 4×100 kbps = **400** kbps.
- Frame rate = input data rate = **100,000** frames/s.
- Frame size = 4 bits (1 from each channel).
- Frame duration: $\frac{1}{100{,}000}$ = **10** μs.
- Verify: $100{,}000 \times 4 = 400{,}000$ bps. ✓

Example 6.10: Framing bits (synchronization bits)

Given: 4 sources, each creating 250 characters/s. 1 synchronization bit added per frame. Interleaved unit = 1 character = 8 bits.

Solution:

1. **Data rate of each source:** 250 chars/s$\times$8 bits/char = **2000** bps = **2** kbps.
2. **Character duration (each source):** $\frac{1}{250}$ = **4** ms.
3. **Frame rate:** Frame rate = character rate = **250** frames/s.
4. **Frame duration:** $\frac{1}{250}$ = **4** ms.
5. **Bits per frame:** 4 sources $\times$ 8 bits + 1 sync bit = **33** bits.
6. **Link data rate:** 250 frames/s $\times$ 33 bits/frame = **8250** bps.

1.3.3 Digital Signal (DS) / T-Line Hierarchy

Telephone companies implement TDM via a standardized hierarchy:

Service	T-Line	Data Rate (Mbps)	Voice Channels
DS-0	—	0.064 (64 kbps)	1
DS-1	T-1	1.544	24
DS-2	T-2	6.312	96
DS-3	T-3	44.736	672
DS-4	T-4	274.176	4032

T-1 Frame Structure Verification

T-1 Frame: 24 channels × 8 bits + 1 framing bit = $\mathbf{24 \times 8 + 1 = 193}$ bits per frame.
Frame rate: Telephone systems sample voice at 8000 Hz (Nyquist for 4-kHz voice bandwidth), so there are **8000** frames/s.
T-1 Data Rate:

$$8000 \text{ frames/s} \times 193 \text{ bits/frame} = \mathbf{1{,}544{,}000} \text{ bps} = \mathbf{1.544} \text{ Mbps}$$

This is the famous T-1 rate!

1.3.4 Statistical TDM

Statistical TDM: Smarter Allocation

Problem with synchronous TDM: Reserved empty slots are wasteful. If a terminal is idle, its slot in every frame is empty.
Statistical TDM solution:

- Slots are **not pre-assigned**. They are allocated *dynamically* — only to sources that have data.
- Empty slots are eliminated; the output frame is **filled with actual data** from active inputs.
- Each slot carries an **address** identifying which channel the data belongs to (since positions are no longer fixed).
- If n sources are active simultaneously, the output rate must accommodate all.
- **Output rate < sum of input rates** in statistical TDM (statistical multiplexing gain), because not all sources are active all the time.

Used in: Internet routers (packet switching is essentially statistical TDM!).

Feature	Synchronous TDM	Statistical TDM
Slot assignment	Fixed per channel	Dynamic (on-demand)
Empty slots	Yes (idle channels waste space)	No (slots always carry data)
Address in slot	Not needed (position = channel)	Required (who does this data belong to?)
Output data rate	= sum of all input rates	< sum of all input rates
Efficiency	Lower	Higher
Complexity	Simple	More complex
Use	T-1 lines	ATM, internet packets

Chapter 2 Spread Spectrum — Section 6.2

Why Spread Spectrum?

Multiplexing (Section 6.1) was about *efficiency*: packing more signals into a link. Spread spectrum is about something different: **security and anti-jamming**.
The core idea: Deliberately *expand* (spread) the bandwidth of a signal far beyond what is strictly needed to carry the data.
Why would anyone spread bandwidth on purpose? Two reasons:

1. **Privacy/Security:** A jammer (enemy, eavesdropper) doesn't know which frequency the signal is using, so jamming or interception is very hard.
2. **Multiple access:** Multiple users can share the same wide bandwidth without interfering, if each uses a unique spreading code (used in CDMA cellular phones).

Requirements:

1. The spread bandwidth B_{SS} must be much greater than the original bandwidth B.
2. The spreading process must be *independent* of the original signal (done via a separate spreading code).

2.1 Frequency Hopping Spread Spectrum (FHSS)

2.1.1 How FHSS Works

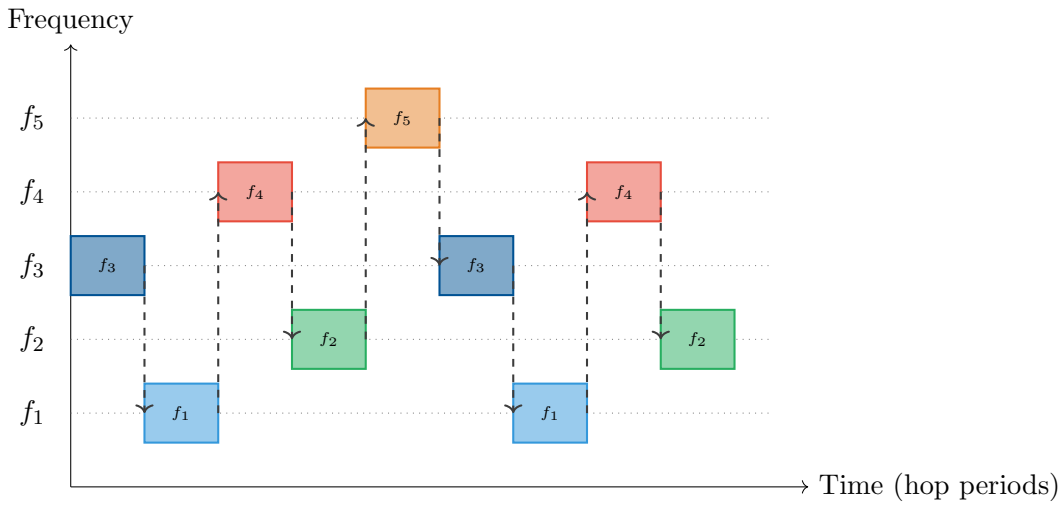
FHSS: Jumping Between Frequencies

In FHSS, the signal doesn't stay on one carrier frequency. Instead, it rapidly **"hops"** from one carrier frequency to another, following a **pseudorandom sequence** known only to the sender and receiver.
Process:

1. The sender has a **frequency table** and a **pseudorandom code generator**. The code determines which frequency to use in each **hop period**.
2. The original signal is modulated onto the selected frequency for one hop

period.

3. After the hop period ends, the sender moves to the next frequency in the sequence.
4. The receiver, using the same code and synchronized with the sender, knows exactly which frequency to tune to at each moment.
5. An eavesdropper doesn't know the hopping sequence, so they can't follow the signal.
6. A jammer can only jam one frequency at a time; since the signal hops away, only a few bits are lost.



Signal hops between frequencies each period
Eavesdropper cannot predict the next frequency

2.1.2 FHSS Bandwidth

FHSS Bandwidth

If the signal hops among k frequencies, and the original signal bandwidth is B:

$$B_{FHSS} \approx k \times B_{hop}$$

Where B_{hop} is the bandwidth at each hopped frequency (approximately equal to the original signal bandwidth B). The total spread bandwidth can be many times the original.

Real-World FHSS Applications:

- **Bluetooth**: Uses 79 frequencies (2402–2480 MHz), hopping 1600 times/second.
- **Early 802.11 Wi-Fi**: Used FHSS in the 2.4 GHz band.
- **Military radios**: FHSS provides anti-jamming capability.

2.2 Direct Sequence Spread Spectrum (DSSS)

2.2.1 How DSSS Works

DSSS: Replacing One Bit with a Chip Sequence

In DSSS, each data bit is replaced with a **sequence of shorter chips**. The chip sequence is called the **spreading code** (or chipping code).
Process:

1. Original data bit (1 or -1 in polar NRZ) is multiplied by a **spreading code** (a sequence of n chips, each +1 or -1).
2. This **multiplication** produces a spread signal. If the bit is 1, the chip sequence is sent unchanged. If the bit is -1, the chip sequence is inverted.
3. The bandwidth increases by a factor of n (the **spreading factor**).
4. At the receiver, the same spreading code is used to **de-spread** the signal and recover the original bits.
5. An eavesdropper without the code sees only noise — they cannot recover the data.

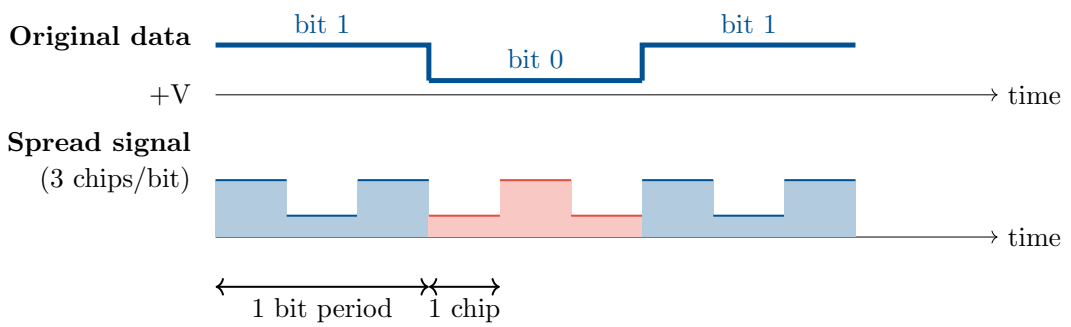
Example: DSSS with Barker Sequence (11 chips)

Given: Original data: bits 1, 0, 1. Barker sequence (11 chips, polar NRZ): $+1-1+1+1-1+1+1+1-1-1-1$ (an 11-chip sequence used in 802.11b Wi-Fi).

Encoding process:

- Bit 1 (= +1): transmitted chips = the Barker sequence itself: $+1-1+1+1-1+1+1+1-1-1-1$
- Bit 0 (= -1): transmitted chips = Barker sequence multiplied by -1: $-1+1-1-1+1-1-1-1+1+1+1$
- Bit 1 (= +1): again the Barker sequence: $+1-1+1+1-1+1+1+1-1-1-1$

Bandwidth change: Original 1 bit → 11 chips. Bandwidth is now 11 times larger!



Bandwidth × 3 (or ×11 for Barker sequence in real DSSS)

2.2.2 Comparison: FHSS vs DSSS

Feature	FHSS	DSSS
Spreading method	Changes carrier frequency	Replaces bits with chip sequence
Bandwidth expansion	Uses multiple channels	Multiplied by spreading factor
Anti-jamming	Excellent (moves away from jammer)	Good (signal looks like noise)
Privacy	Good	Good
Multiple access	Users use different hop sequences	Users use different codes (CDMA)
Used in	Bluetooth, early Wi-Fi	802.11b Wi-Fi, GPS, CDMA cellular

Chapter Summary

Key Takeaways — Chapter 6

Section 6.1 — Multiplexing:

- **FDM**: Analog signals share bandwidth by using different frequency bands. Guard bands prevent interference.
- Example: AM radio (10 kHz each), FM (200 kHz each), TV (6 MHz each), AMPS (30 kHz each).
- **WDM**: Like FDM but for light in optical fiber. Used in backbone internet.
- **Synchronous TDM**: Digital signals share time. Fixed slots per channel. Output rate = $n\times$ input rate. T-1 = 1.544 Mbps (24 channels $\times$ 64 kbps + framing).
- **Statistical TDM**: Dynamic slot allocation; no empty slots; needs address field; more efficient.

Section 6.2 — Spread Spectrum:

- **FHSS**: Signal hops among many frequencies using a pseudorandom sequence. Anti-jamming, privacy.
- **DSSS**: Each bit replaced by a chip sequence; bandwidth expands by spreading factor. 802.11b uses 11-chip Barker sequence.

Critical Formulas:

- FDM total bandwidth = $\sum$ channel bandwidths + $(n-1)\times$ guard band
- TDM output rate = $n\times$ input rate (synchronous)
- T-1 frame = $24 \times 8 + 1 = 193$ bits; 8000 frames/s = 1.544 Mbps